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A gated recurrent unit neural networks based wind speed error correction model for short-term wind power forecasting

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ABSTRACT

With the growing penetration of wind power, the wind power forecasting is fundamental in aiding the grid scheduling and electricity trading. In this paper, a numerical weather prediction wind speed error correction model based on gated recurrent unit neural networks is proposed for short-term wind power forecasting. Firstly, the standard deviation of numerical weather prediction wind speed error is extracted as weights, and these weights are rearranged according to the numerical weather prediction wind speed time series to get the weight time series. Then, the bidirectional gated recurrent unit neural networks based error correction model is proposed to correct error of numerical weather prediction wind speed with the inputs as numerical weather prediction wind speed, trend and detail terms of the weight time series. The wind power curve model is applied to forecast short-term wind power by using corrected numerical weather prediction wind speed. Finally, the effectiveness of the proposed method is compared with benchmark models by using actual data of wind farm, and the results show that the proposed model outperforms these benchmark models.

1. Introduction

Wind power has become the second largest renewable energy source [1] and is widely used in many areas [2]. However, wind energy has the characteristics of intermittency and volatility [3], which bring a big challenge for grid scheduling and electricity trading. In the growing wind penetration scenario, accurate wind power forecasting is vital for mitigating the undesirable effects from these characteristics.

According to time-scales, wind power forecasting can be classified into four categories: ultra-short-term from minutes to hours, short-term from hours to days, medium-term from weeks to months, and long-term from months to years [4]. The ultra-short-term and short-term wind power forecasting provide the basis for grid scheduling and electricity trading, and the medium-term and long-term wind power forecasting are significant for site selection, performance forecasting, planning of windmills, and the selection of an optimal size of the wind machine for particular site [5]. This paper focus on the short-term wind power forecasting that is 24 h in advance to assist grid scheduling and electricity trading.

Among the proposed wind power forecasting methods, two types of data are utilized to build wind power forecasting models. One is the measured wind power data, another is the numerical weather prediction (NWP) data [6]. The measured wind power data is mostly used for the ultra-short-term wind power forecasting. Because wind power is directly affected by cubic of wind speed, NWP data is an important data source for wind power forecasting. The NWP wind speed is mainly used for short-term wind power forecasting models. The existing methods of wind power forecasting by using NWP data can be classified into statistical methods, intelligent learning methods, and hybrid methods [7,8]. However, the intermittency and volatility of wind speed leads to the uncertainty of NWP. The uncertainty of NWP affects the accuracy of wind power forecasting. This brings great challenges for wind power forecasting. So, data mining methods are used to forecast short-term wind power forecasting by using NWP [9], such as artificial neural network (ANN) combined with computational fluid dynamics [10], daily similarity cluster [11], Hilbert-Huang transform [12], and principal component analysis [13]. Besides, Gaussian Processes [14,15] and Sparse Bayesian Learning [16] are utilized to built wind power forecasting model. In [17], a boundary layer scaling method is used to forecast long-term average near-surface wind power. However, there is always error existing in NWP wind speed, so extended numerical weather variables

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[18] and bias correction methods [19] are utilized to correct NWP wind speed in wind power forecasting.

In order to improve the accuracy of wind power forecasting, many error correction models (ECM) are proposed [20–25]. However, these error correction models are built to provide point to point forecasting without using time series characteristics of NWP and wind power. In [26], the identification method and correction method for time series power are proposed to improve the cost and curtailment benefits. Further, the vector error correction models are proposed to correct time series error [27–29]. So, time series characteristics of NWP wind speed error can be applied to build an error correction model to improve accuracy.

As time series model is affected by the time series features, extraction of time series features becomes very important. Extraction methods of time series features are widely studied, such as time Series-Histogram of Features method [30], temporal importance curve [31], spatial assembling distance [32], wavelet principal component analysis [33], weighted shape-based time-series clustering [34], similar shape functional time series predictor [35], supervised aggregative feature extraction method [36], shape-aware time series matching algorithm [37]. However, these literatures rarely use statistics of time series.

According to the discussion above, the existing statistical model or time series model don't make full use of statistical characteristic and time series feature to correct NWP wind speed. The Gated recurrent unit neural networks (GRUNNs) is introduced to make full use of both statistical features and dynamic temporal behavior of the time series. The RNN is used for typical time series in different fields [38–40]. Gated recurrent unit (GRU) [41], a RNN variant, is regarded as a simple version of long short-term memory [42]. The biggest difference between GRUNN and traditional neural network is that GRU is introduced to overcome some drawbacks. The structure of GRU effectively mitigates the vanishing gradient problem [43]. Further, GRU is a popular time series forecasting model and expertly deals with long-term data dependency. So, GRUNNs has the ability to correct error of NWP wind speed.

The paper builds a wind speed error correction model by using GRUNN based both on statistical characteristic and time series characteristic. There are two main contributions of this paper. First, weight time series with both statistical characteristic and time series characteristic of data is presented as an input to wind speed error correction model. Then, the bidirectional gated recurrent unit neural networks (GRUNNs) based NWP wind speed error correction model is proposed to correct the NWP wind speed with considering both the statistical characteristic and the time series characteristic.

This paper is organized as follows. Section 2 describes the existing problem, features analysis and forecasting framework. Section 3 introduces the bidirectional GRUNNs based NWP wind speed error correction model. Section 4 discusses the design and results of the experiment. Finally, Section 5 summarizes the paper.

2. Problem description and forecasting framework

In this section, the problem of wind power forecasting by using NWP wind speed is described. Then, the features of NWP wind speed is analyzed. Further, the wind power forecasting framework is proposed.

2.1. Problem description

In nature, the wind speed is a continuous physical quantity. The anemometer tower measures the wind speed at a time interval, like 1 second to obtain wind speed time series. Further, the wind speed time series with a larger time interval such as 15 min are obtained by averaging the measured wind speed.

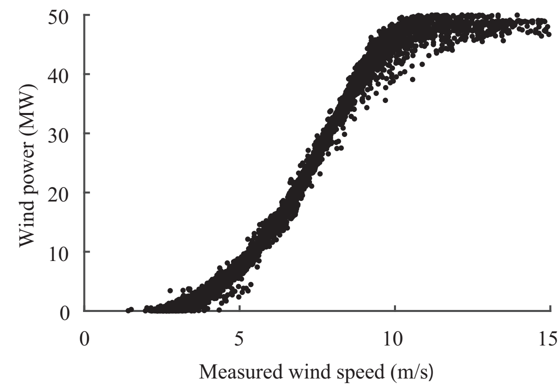

Fig. 1. Relation between measured wind speed and wind power in an actual wind farm.

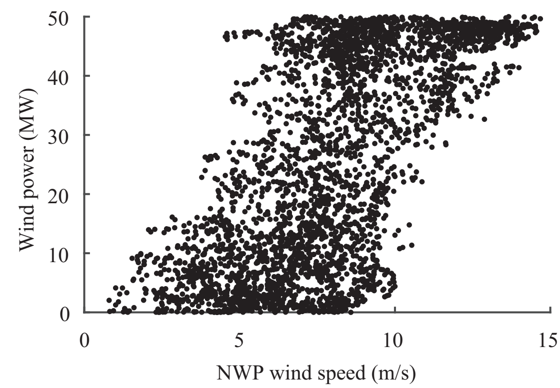

Fig. 2. Relation between NWP wind speed and wind farm output power.

A wind turbine captures the wind energy and converts the energy into electrical power. The output power of a wind turbine is related to the kinetic energy of the air passing through the effective area of the blades. The output electrical power is given as follows:

$$P_m = \frac{1}{2} C_P \rho A V^3 \tag{1}$$

where P_m is the output power of wind turbine, C_P is the turbine power coefficient, ρ is the air density, A is the area swept by the turbine blades, V is the wind speed. In Fig. 1, the relationship between the measured wind speed and the wind power of an actual wind farm in Sichuan Province in China meets the wind-power curve of the wind turbine.

Due to the uncertainty of NWP, the NWP wind speed has error ε. Then, the Eq. (1) is changed as follows:

$$P_m = \frac{1}{2} C_P \rho A (V + \varepsilon)^3 \tag{2}$$

The NWP wind speed can be used to forecast wind power. In the actual wind farm, the relation between NWP wind speed and wind power is shown in Fig. 2.

Figs. 1 and 2 show that the wind speed error of NWP will lead to forecasting error. Error correction method should be developed to reduce the wind speed error of NWP so as to improve the accuracy of wind power forecasting. So, the features of NWP wind speed should be analyzed and extracted.

2.2. Features analysis of NWP wind speed

To investigate the statistical characteristics of wind speed time series, the histogram of measured wind speed and NWP wind

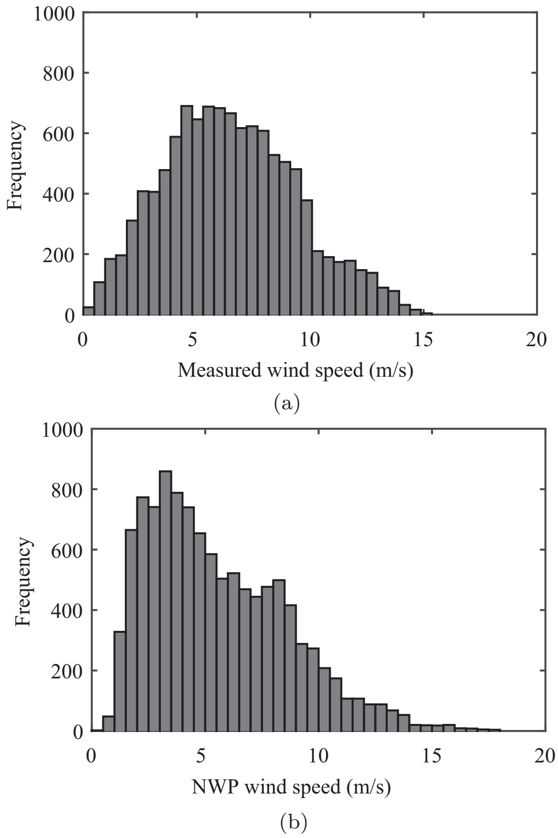

Fig. 3. (a) Histogram of mearsred wind speed (b) Histogram of NWP wind speed.

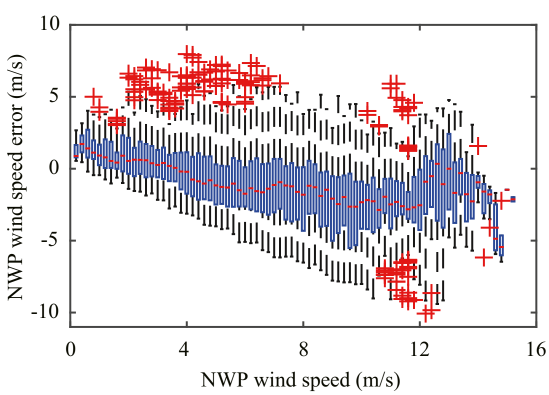

Fig. 4. Boxplot of wind speed error.

speed are shown in Fig. 3(a) and (b). The distribution of NWP wind speed is different from the distribution of measured wind speed.

Assuming the measured wind speed is accurate, the NWP wind speed error is calculated by subtracting NWP wind speed from the measured wind speed. In Fig. 4, the black line shows the range of NWP wind speed error under different NWP wind speed, the red dot represents medium NWP wind speed errors under different NWP wind speed,the red cross represents the outlier. Each value of NWP wind speed has its corresponding distribution of NWP wind speed error.

2.3. Forecasting framework

From the analysis above, an error correction model of NWP wind speed based on GRUNNs is proposed for NWP wind speed

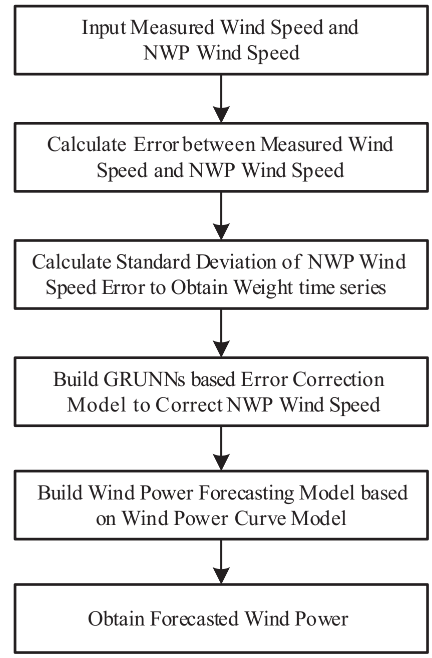

Fig. 5. Forecasting frame of wind power.

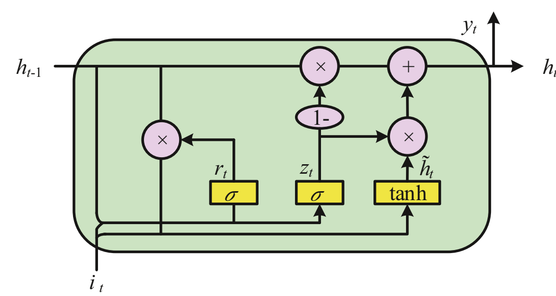

Fig. 6. Architecture of GRU.

error correction to improve accuracy of short-term wind power forecasting. The forecasting framework is shown in Fig. 5.

The standard deviation of NWP wind speed error is set as weight time series. The weight time series is decomposed into trend terms and detail terms that are input into GRUNNs based error correction model to correct NWP wind speed. The corrected NWP wind speed is applied to forecast wind power to improve forecast accuracy.

3. Error correction model of NWP wind speed

In this section, the error correction model of NWP wind speed based on bidirectional GRUNNs is proposed to correct NWP wind speed error.

3.1. Bidirectional GRUNNs

The architecture of GRU is shown in Fig. 6, there are two gates in GRU. A reset gate r adjusts the incorporation of new input with the previous memory and an update gate z controls the preservation of the previous memory. It can be described as

$$\begin{cases} r_t = \sigma\left(i_t W_{xr} + h_{t-1} W_{hr} + b_r\right) \\ z_t = \sigma\left(i_t W_{xz} + h_{t-1} W_{hz} + b_z\right) \\ \tilde{h}_t = \tanh\left(i_t W_{xh} + r_t \odot h_{t-1} W_{hh} + b_h\right) \\ h_t = z_t \odot h_{t-1} + (1 - z_t) \odot \tilde{h}_t \end{cases} \tag{3}$$

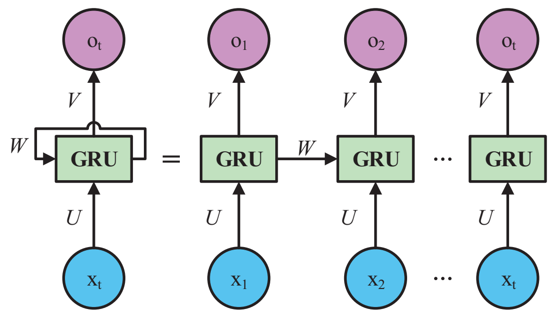

Fig. 7. Architecture of GRU-based RNN.

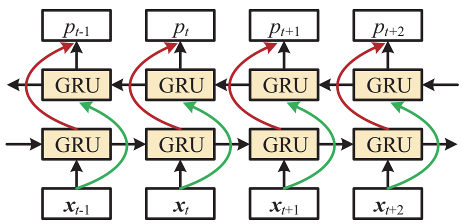

Fig. 8. Structure of bidirectional GRUNNs.

Where i_t is the current time input, h_{t-1} is the last time output, $\tilde{h}_t$ and y_t are current time output that pass to next cell and next layer, respectively. W_{xr}, W_{xz} and W_{xh} are three parameter matrices of current time input i_t to r_t, z_t and $\tilde{h}_t$ that are trained by neural network. W_{hr}, W_{hz} and W_{hh} are three parameter matrices of last time output h_{t-1} to r_t, z_t and $\tilde{h}_t$ that are trained by neural network.

In the RNN structure, when the neuron is designed as a GRU, the new architecture of GRU-based RNN is shown in Fig. 7. The parameters of RNN are trained and calculated by the backpropagation network. It can be described as

$$o_t = Vf(Ux_t + Wf(Ux_{t-1} + Wf(Ux_{t-2} + Wf(Ux_{t-3} + \ldots)))) \quad (4)$$

where o_t is the result of one neural cell. x_t is the input of t moment. U, V and W are weight matrices of x, h and output layer. $f(\cdot)$ is the activation function.

For the GRU-based RNN, if the independent GRUNNs are connected as shown in Fig. 8, the structure is developed as bidirectional GRUNNs. This structure enables GRUNNs to process the sequence inputs in two directions including forward and backward ways with two individual hidden layers. Each hidden layer can captures past (forward direction) and future (backward direction) data information at the same time step. Bidirectionality of this structure increases the model capacity and flexibility.

3.2. Error correction model based on GRUNNs

Based on the structure of bidirectional GRUNNs, the wind speed error correction model is proposed and can be described as Fig. 9. Inputs are time-series data denoted as $\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_n]$, where n is the length of training data. Local features are extracted from inputs of training data.

3.2.1. Local-feature extraction

Each input is divided into multiple local segments. Each segment is a window of original signal with a length l. For example, the jth local window is a segment starting from time step $\boldsymbol{x}_j$ to $\boldsymbol{x}_{j+l-1}$. We assume that after local-feature extraction, the original inputs are transformed to a sequence of local features as

$$\boldsymbol{X}_t = [\boldsymbol{x}_t,\ \boldsymbol{x}_{t+1},\ \cdots,\ \boldsymbol{x}_{t+l-1}]^{\mathrm{T}},\ t \in 1,\ 2,\ \cdots,\ n-l+1 \quad (5)$$

where $\boldsymbol{X}_t$ consists of l features extracted from the tth local window. Compared with the original time series $\boldsymbol{X}$, the local-feature sequence is much shorter and conveys more discriminative information than the noisy original inputs.

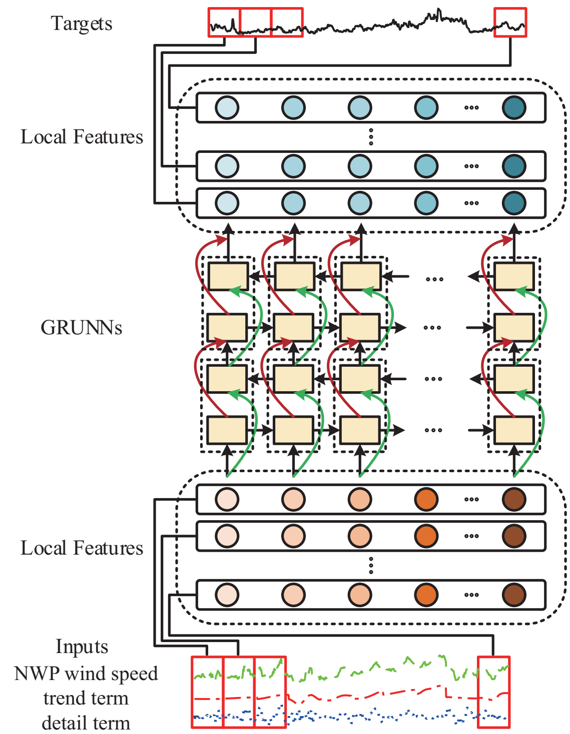

Fig. 9. Architecture of wind speed error correction model.

3.2.2. Process of modeling

In order to utilize the features of wind speed time series, the weight time series are decomposed into the trend terms and detail terms by empirical mode decomposition. NWP wind speed is v, trend term is c_t, and details term is c_d. In the proposed error correction model, the input matrix $\boldsymbol{X}\{x_1, x_2, \cdots, x_n\}$ is shown in Eq. (6). The target matrix is $\boldsymbol{P}\{p_1, p_2, \cdots, p_n\}$.

$$\boldsymbol{X} = \begin{bmatrix} \boldsymbol{x}_1 \\ \boldsymbol{x}_2 \\ \vdots \\ \boldsymbol{x}_n \end{bmatrix} = \begin{bmatrix} v_1 & c_{t1} & c_{d1} \\ v_2 & c_{t2} & c_{d2} \\ \vdots & \vdots & \vdots \\ v_n & c_{tn} & c_{dn} \end{bmatrix} \quad (6)$$

By using Eq. (6), the local-feature matrix are obtained.

$$\boldsymbol{X}_t = [\boldsymbol{x}_t\ \boldsymbol{x}_{t+1}\ \cdots\ \boldsymbol{x}_{t+l-1}]^{\mathrm{T}},\ t \in 1,\ 2,\ \cdots,\ n-l+1 \quad (7)$$

$$\boldsymbol{P}_t = [p_t\ p_{t+1}\ \cdots\ p_{t+l-1}]^{\mathrm{T}},\ t \in 1,\ 2,\ \cdots,\ n-l+1 \quad (8)$$

The error correction model is built by bidirectional GRUNNs.

$$[\boldsymbol{P}_1\ \boldsymbol{P}_2\ \cdots\ \boldsymbol{P}_{n-l+1}] = f([\mathbf{X}_1\ \mathbf{X}_2\ \cdots\ \mathbf{X}_{n-l+1}]) \quad (9)$$

where $f(\cdot)$ is training function of bidirectional GRUNNs and the whole process can be explained by (3) and (4). The Eq. (10) can be rewritten as

$$\begin{bmatrix} p_1 & p_2 & \cdots & p_{n-l+1} \\ p_2 & p_3 & \cdots & p_{n-l+2} \\ \vdots & \vdots & \ddots & \vdots \\ p_l & p_{l+1} & \cdots & p_n \end{bmatrix} = f\left(\begin{bmatrix} \boldsymbol{x}_1 & \boldsymbol{x}_2 & \cdots & \boldsymbol{x}_{n-l+1} \\ \boldsymbol{x}_2 & \boldsymbol{x}_3 & \cdots & \boldsymbol{x}_{n-l+2} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{x}_l & \boldsymbol{x}_{l+1} & \cdots & \boldsymbol{x}_n \end{bmatrix}\right) \quad (10)$$

4. Experiment results and analysis

In this section, an experiment is designed to validate the proposed method. Actual data sets of wind farms are used to com-

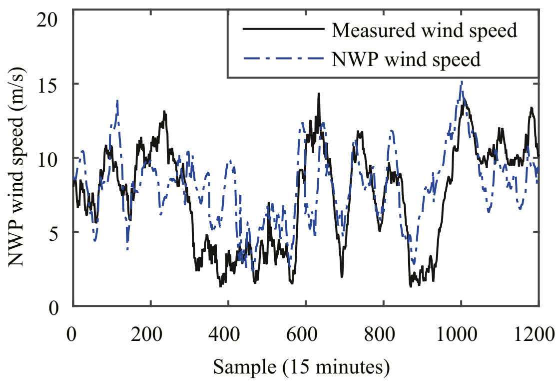

Fig. 10. Measured wind speed and the NWP wind speed.

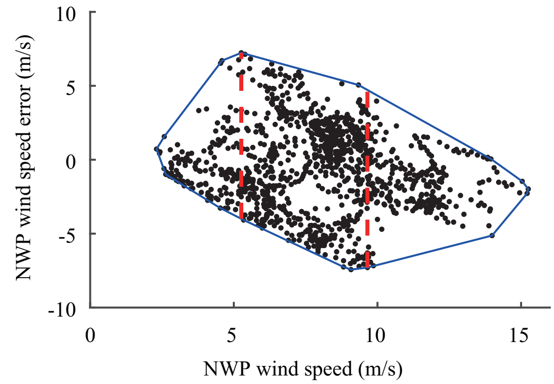

Fig. 11. Distribution of wind speed error.

pare the performance of the proposed model with the benchmark models.

4.1. Experiment design

All of the experiments were done on a Windows 10 PC with a Intel(R) Xeon(R) E5-2620 v3 2.4 GHz CPU, 64GB of memory, NVIDIA GeForce GTX 1080 Ti, and Python version 3.6 with TensorFlow 1.8.0.

A set of wind speed, wind power and NWP wind speed data, obtained from the actual data of a wind farm located in the Sichuan Province, China in 2016, sampled at a period of 15 min, was used. The proposed method was compared with SVM and ANN benchmark models [13,44] to verify the effectiveness.

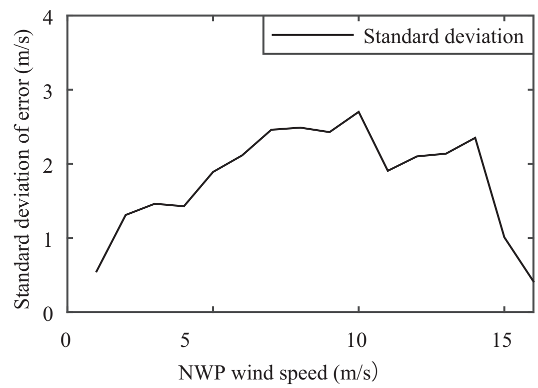

Fig. 12. Standard deviation of NWP wind speed error.

4.2. Evaluation criteria

To evaluate the forecasting performance of the proposed model, root mean square error (RMSE) and mean absolute error (MAE) of forecasting wind power are employed as the criteria. Within the range [0, 1], the smaller RMSE and MAE means that the performance of the proposed model is better. The equations are expressed as follow:

$$RMSE = \sqrt{\frac{1}{l}\sum_{i=1}^{l}\left(\frac{P_i - \hat{P}_i}{Cap_i}\right)^2} \tag{11}$$

$$MAE = \frac{1}{l}\sum_{i=1}^{l}\left(\frac{\left|P_i - \hat{P}_i\right|}{Cap_i}\right) \tag{12}$$

where l is the number of samples, P_i is ith actual power, $\hat{P}_i$ is ith forecasting wind power, Cap_i is ith installed capacity of wind farm.

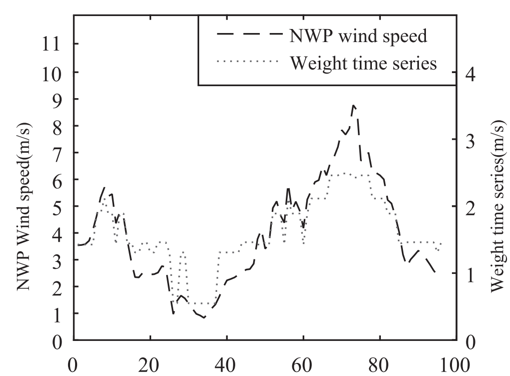

Fig. 13. Weigh time series of NWP wind speed time series.

4.3. Experiment results

The experiments using the actual data of wind farm are carried out to evaluate the performance. The generation process of the weight time series is explained. Then, the forecasting results are illustrated.

4.3.1. Weight time series

For the data set of 1200 sample points, the measured wind speed and the NWP wind speed are shown in Fig. 10. NWP wind speed error is calculated by subtracting NWP wind speed from measured wind speed. The distribution between NWP wind speed and NWP wind speed error is shown in Fig. 11. The deviation between NWP wind speed and measured wind speed is changing as NWP wind speed increases. The range of the error increases first then decreases with the increase of NWP wind speed. The range of NWP wind speed error in middle NWP wind speed from 5 to 10 m/s is larger than others. The standard deviation of NWP wind speed error is shown in Fig. 12. The standard deviations of NWP wind speed error are extracted as weights.

According to the NWP wind speed time series, weights in Fig. 12 are rearranged to get the weight time series in Fig. 13. It shows that not only the NWP wind speed has the characteristics of time series, but also the error of NWP wind speeds has the characteristics of time series. For example, at the adjacent points of 72nd sample points, the NWP wind speed and NWP wind speed error are larger than other points, such as point of 30th and 95th sample points. Take other periods for example, the NWP wind speed and NWP wind speed error are shown as Fig. 14. This characteristic requires to consider forward and backward time

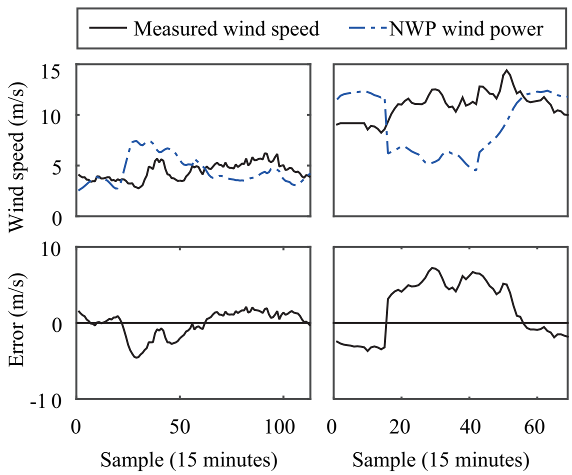

Fig. 14. Wind speed data and errors of two data sets.

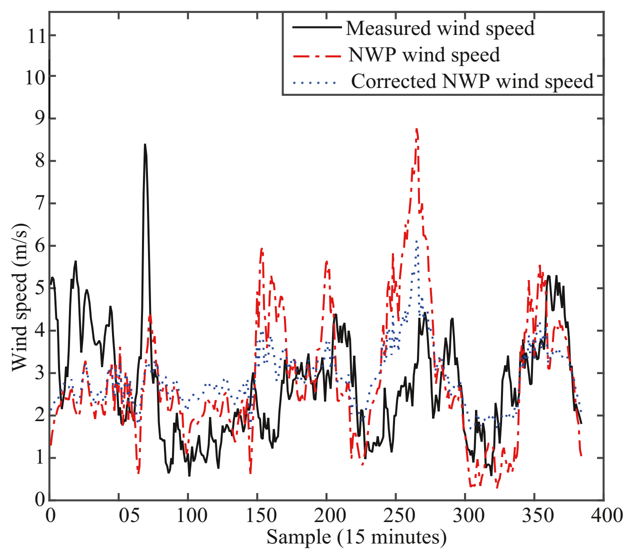

Fig. 15. Corrected NWP wind speed.

series information. So, the time series characteristics of NWP wind speed error is utilized to correct NWP wind speed.

4.3.2. Analysis and comparison of forecasting results

Simulations are carried out to evaluate the performance of the proposed method. The standard deviations of NWP wind speed

Table 1
RMSE of forecasting wind power.

Days	NWP wind speed	SVM	ANN	Proposed model
1	14.04%	13.51%	13.50%	13.45%
2	21.35%	8.28%	8.26%	6.01%
3	4.24%	3.00%	3.06%	2.40%
4	10.26%	13.65%	13.37%	12.14%
5	11.43%	12.29%	12.33%	12.07%
6	12.89%	12.17%	12.21%	12.11%
7	12.14%	9.74%	9.79%	9.69%
8	11.66%	5.06%	4.94%	4.76%
9	38.46%	15.46%	15.56%	15.23%
10	37.49%	14.55%	13.69%	13.49%
11	15.08%	18.64%	18.64%	16.89%
12	40.39%	16.74%	16.63%	15.49%
13	14.14%	17.97%	17.96%	14.88%
14	37.24%	23.65%	23.77%	19.34%
15	10.97%	7.96%	7.97%	6.99%
16	31.18%	21.62%	21.49%	18.26%
17	50.24%	21.60%	21.63%	18.58%
18	19.39%	10.54%	10.34%	10.01%
19	21.60%	16.21%	16.53%	15.35%
20	2.80%	8.77%	8.73%	8.01%
21	19.96%	7.46%	6.97%	7.12%
22	37.82%	15.03%	14.82%	13.87%

error are extracted as weights in Fig. 12. The standard deviation of NWP wind speed error time series is rearranged to get the weight time series. The weight time series is decomposed into trend terms and detail terms by empirical mode decomposition. NWP wind speed, trend terms and detail terms of weight time series are used as inputs of correction model in (6) and the historical measured wind speed is used as the output to train the model in (8) and (10). Then, the proposed correction model is obtained. Because of the cubic relationship between wind power and wind speed, the inaccurate NWP wind speed in high wind speed segment causes more error than low wind speed segment. Fig. 15 shows that the proposed method performs well when NWP wind speed is much higher than the measured wind speed. Thus, the proposed method is very meaningful.

This corrected NWP wind speed is used to forecast wind power by using the wind power curve. The forecasted wind power of the proposed method is compared with the ANN and SVM models. The wind power forecasting results of these cases are shown in the Fig. 16. The RMSE and MAE of forecasting results are shown in Tables 1 and 2. From these tables, the accuracy of wind power forecasting has a great improvement by using the corrected NWP wind power. The proposed model outperforms than all the benchmark models. The experiments validate the effectiveness of the proposed method.

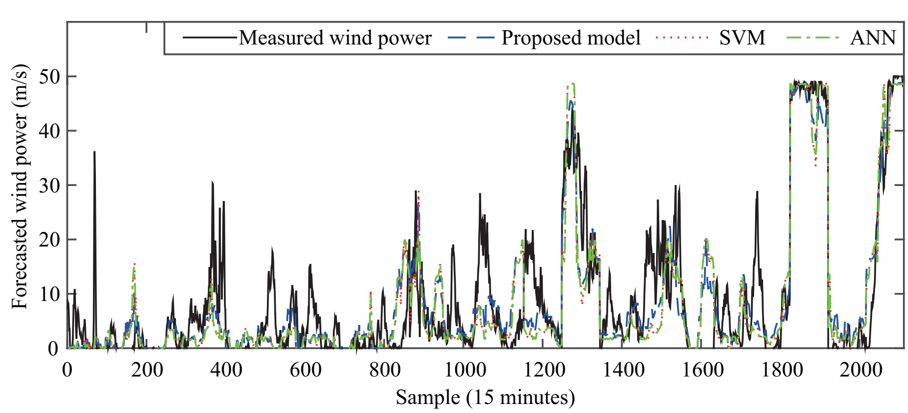

Fig. 16. Wind power forecasting result of proposed model and benchmark model.

Table 2
MAE of forecasting wind power.

Days	NWP wind speed	SVM	ANN	Proposed model
1	7.42%	6.90%	6.80%	6.87%
2	13.65%	5.21%	5.23%	4.44%
3	2.81%	1.84%	1.85%	1.49%
4	8.19%	9.93%	9.88%	8.65%
5	8.04%	6.29%	6.23%	6.16%
6	9.64%	8.35%	8.43%	8.76%
7	8.58%	6.69%	6.78%	6.89%
8	7.68%	3.52%	3.47%	3.70%
9	28.72%	11.81%	11.90%	11.91%
10	28.77%	10.35%	10.35%	10.02%
11	11.38%	12.77%	12.76%	12.12%
12	29.17%	12.56%	12.40%	12.40%
13	11.70%	13.23%	13.23%	10.49%
14	31.61%	18.94%	19.02%	14.41%
15	8.53%	5.99%	6.00%	5.47%
16	22.14%	18.14%	18.07%	15.20%
17	35.67%	14.97%	14.95%	13.77%
18	13.85%	7.99%	7.62%	7.74%
19	16.37%	11.17%	11.29%	10.97%
20	1.94%	4.79%	4.83%	6.07%
21	14.27%	5.06%	4.82%	6.12%
22	27.41%	10.92%	10.63%	10.02%

5. Conclusion

In this paper, an NWP wind speed error correction model based on GRUNNs is proposed for short-term wind power forecasting. Firstly, the feature of NWP wind speed is analyzed, and the standard deviation of NWP wind speed error is extracted as weights of NWP wind speed time series. Then, the bidirectional GRUNNs based error correction model is proposed to correct NWP wind speed. Finally, the effectiveness of the proposed forecasting model is compared with the benchmark models. From the experiment results, the proposed model showed the better capability of short-term wind power forecasting. This method can also be used in medium-term or long-term wind power forecasting. Because the medium-term and long-term numerical weather forecasting data have both the statistical characteristic and the time series characteristic, this method can also be used in medium-term or long-term wind power forecasting.

Declaration of Competing Interest

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

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